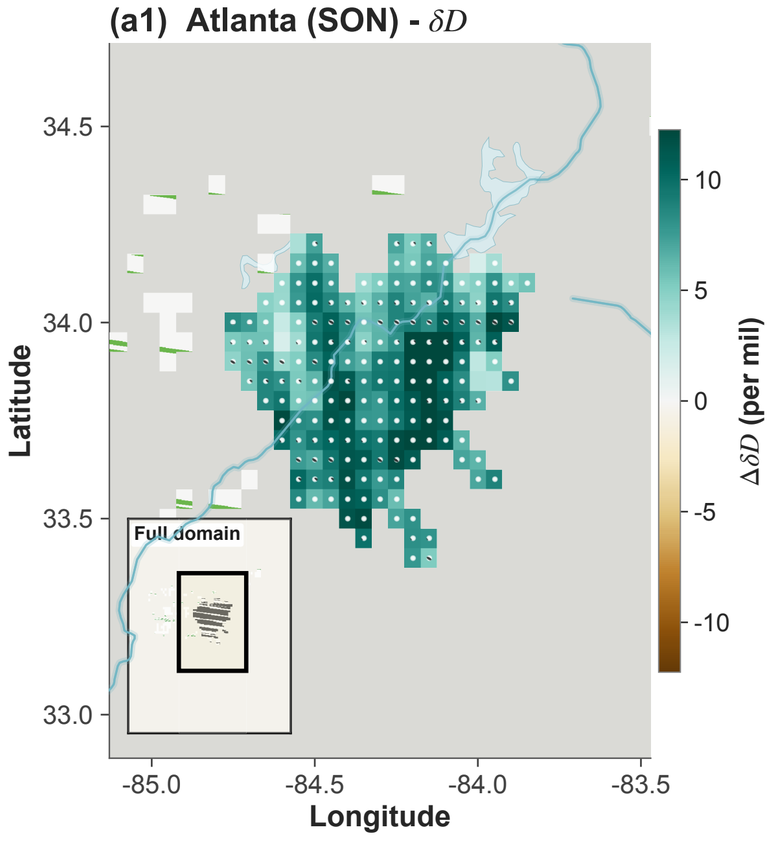
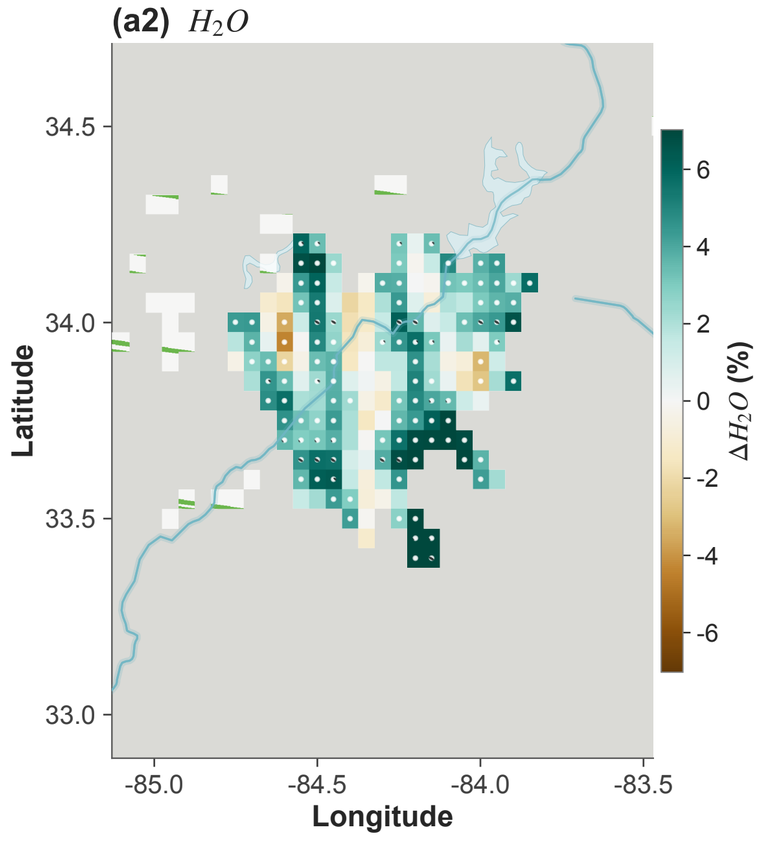
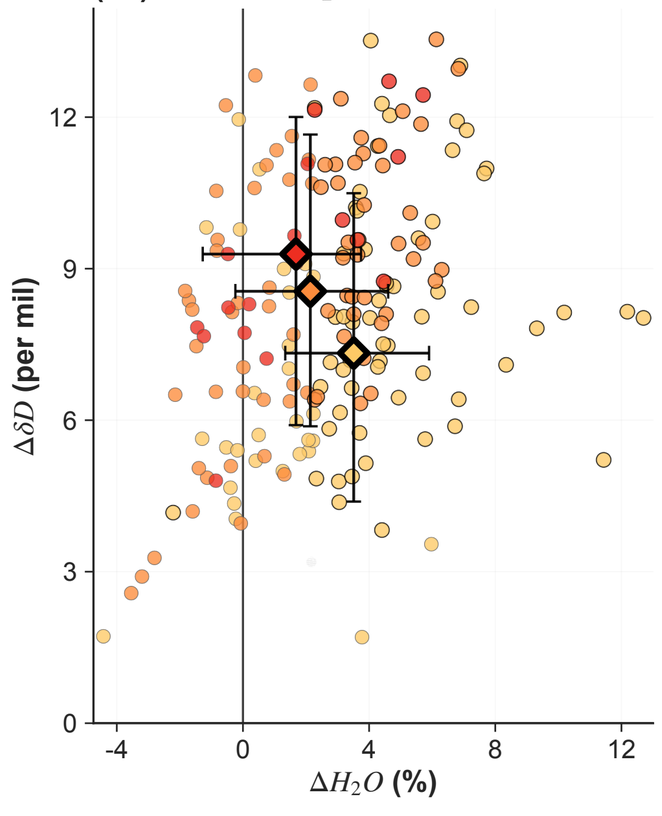
(a1) Atlanta (SON) - δD



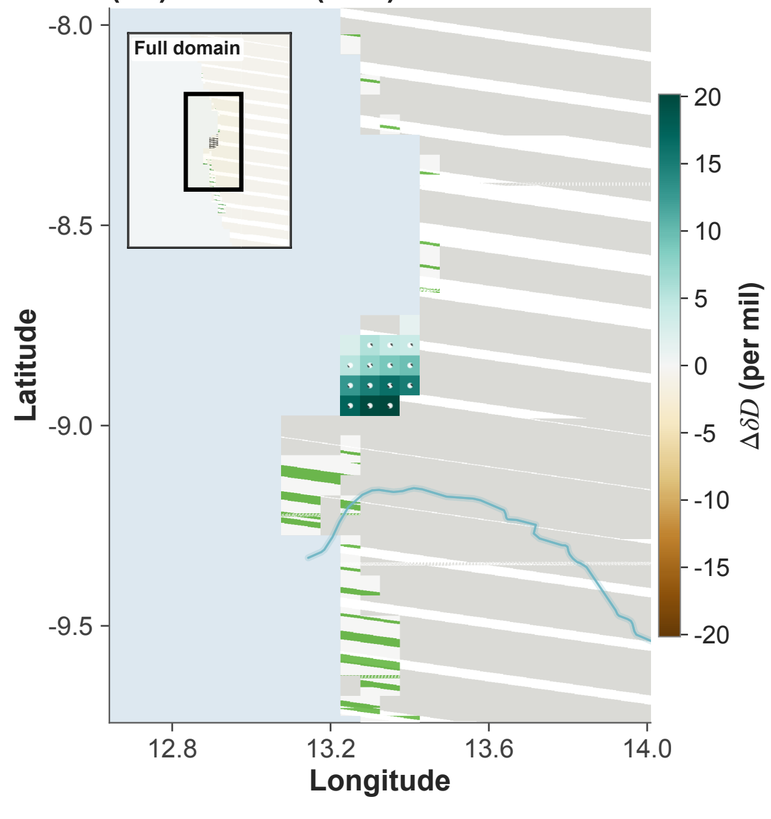
(a2) H_2O



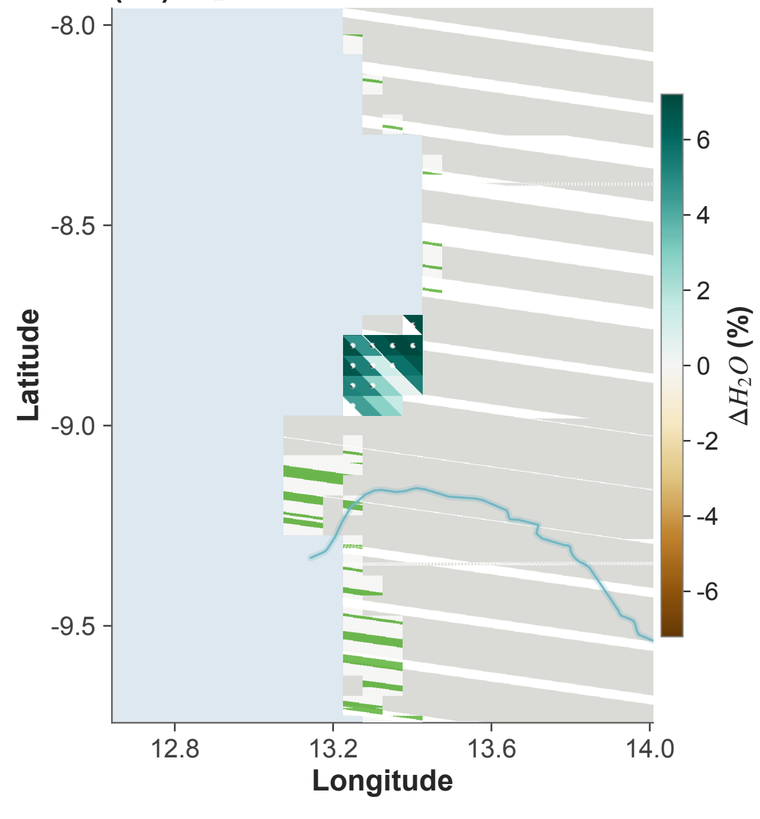
(a3) Paired ΔH_2O-$\Delta\delta D$



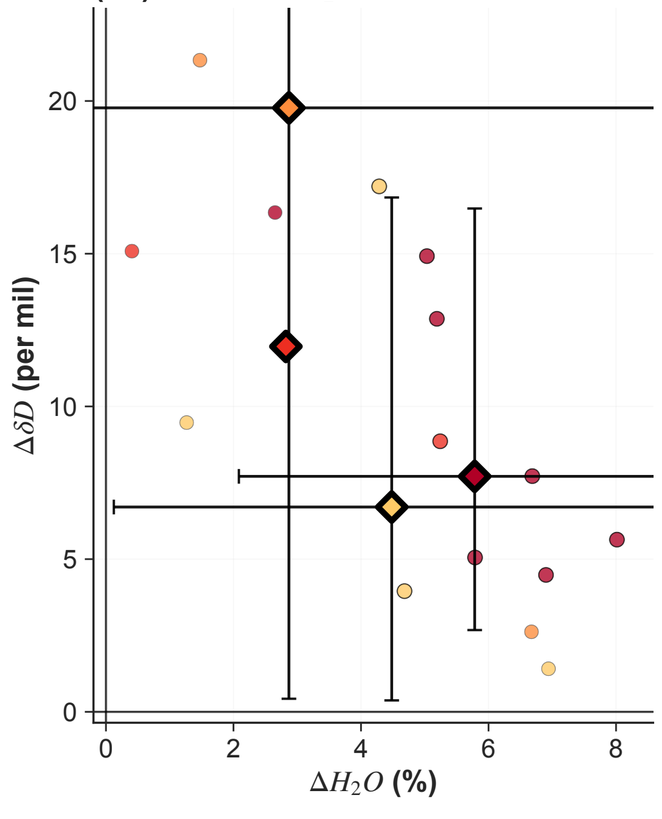
(b1) Luanda (DJF) - δD



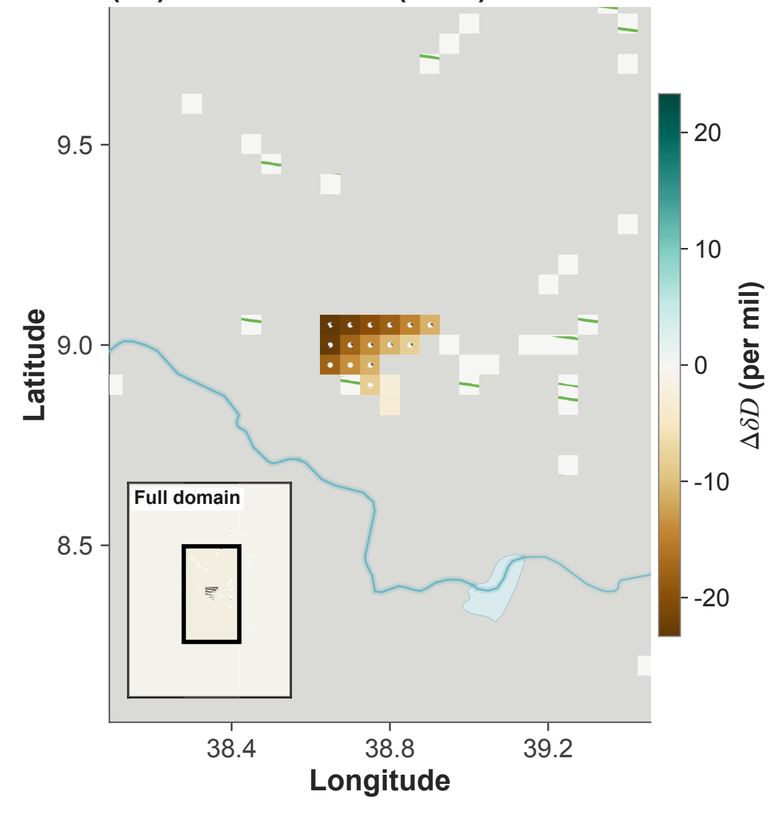
(b2) H_2O



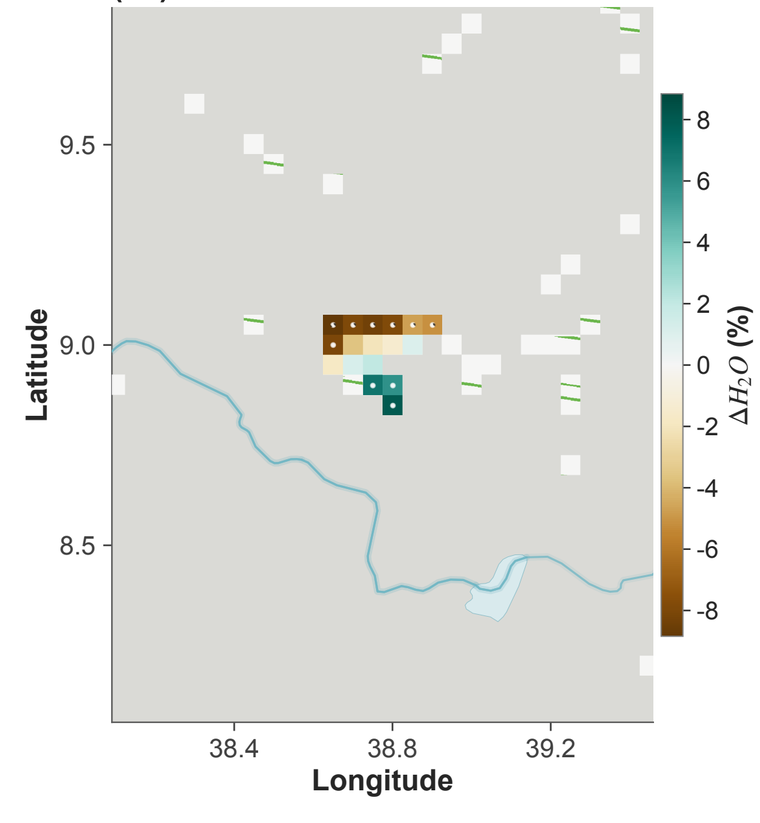
(b3) Paired ΔH_2O-$\Delta\delta D$



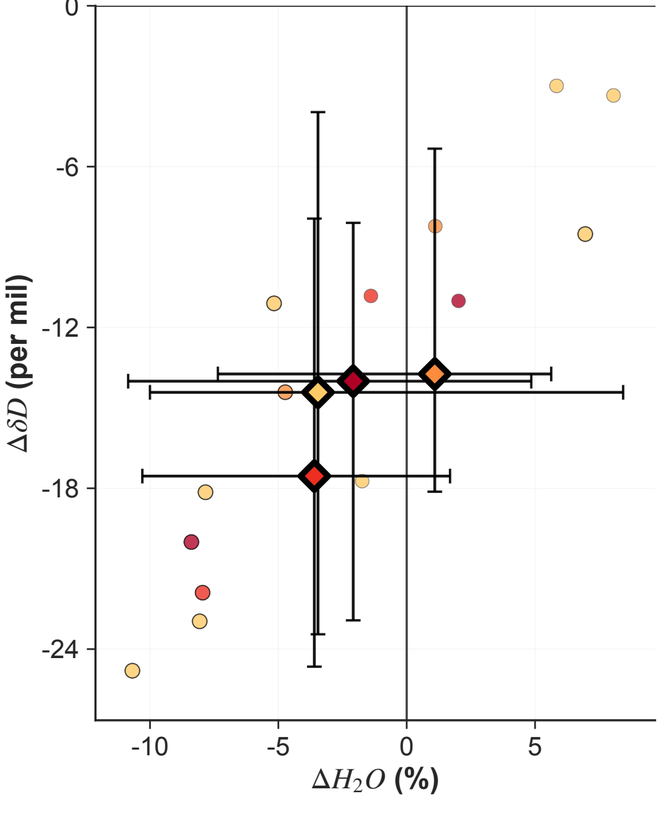
(c1) Addis Ababa (SON) - δD



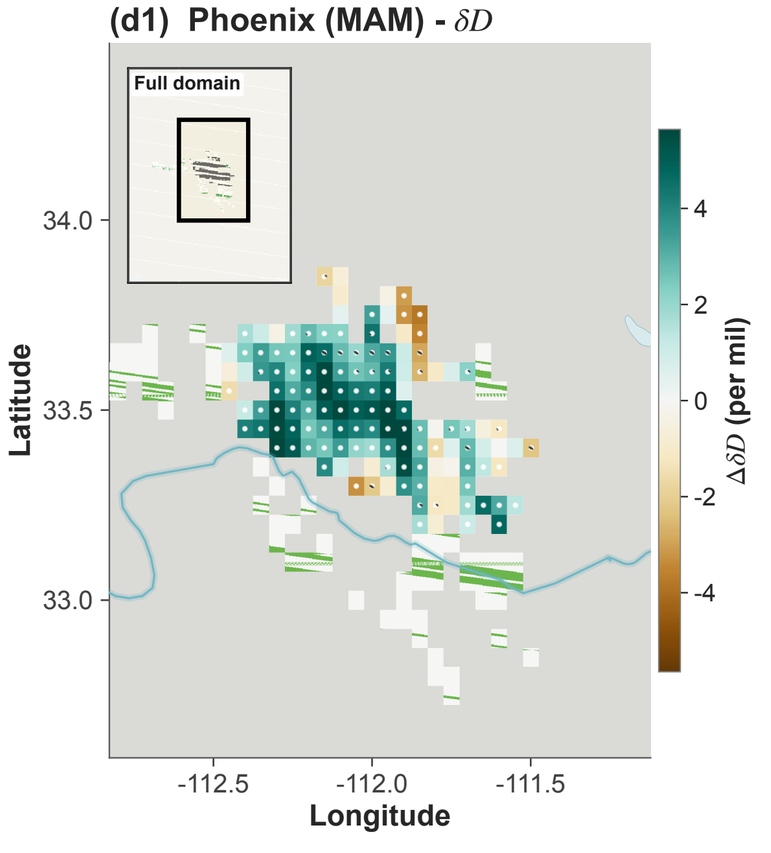
(c2) H_2O



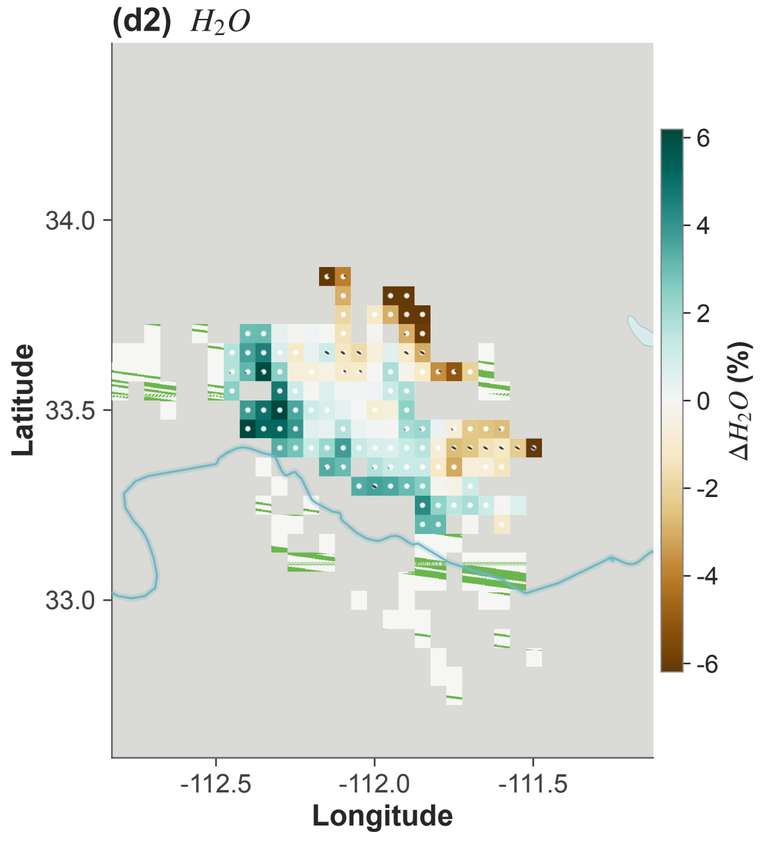
(c3) Paired ΔH_2O-$\Delta\delta D$



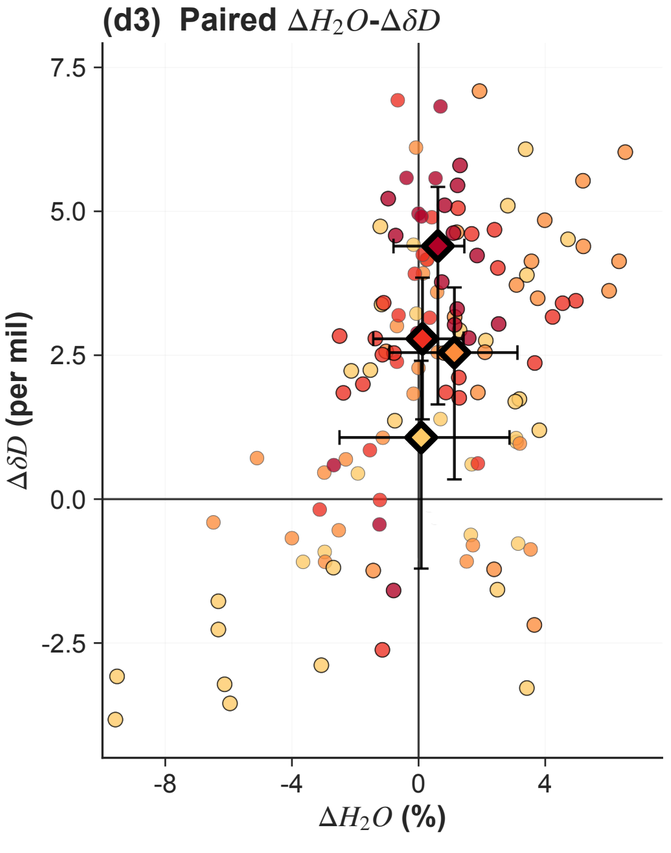
(d1) Phoenix (MAM) - δD



(d2) H_2O



(d3) Paired ΔH_2O-$\Delta\delta D$



Control pixels • Map stippling: $|\text{anomaly}| \geq 2SE$ Urban settlement: 20-40% 40-60% 60-80% 80-100% bin median + 95% spatial-block interval (25 km; pixel fallback if <5 blocks)